NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – I : PRINCIPLES OF SOLAR RADIATION

I. Role & Potential of New and Renewable Sources

Energy sources are of two types: **Conventional** (coal, oil, gas, large hydro, nuclear) – limited stock, polluting; and **Non-conventional / renewable** (solar, wind, biomass, geothermal, ocean, small hydro) – replenished naturally, almost no pollution.

❖Why do we need renewable energy?

- Fossil fuel reserves are **finite** – coal ~100-150 yrs, oil/gas ~40-60 yrs at present rate.
- Burning fossil fuels causes CO_2, SO_2, NO_x emission → global warming, acid rain, smog.
- Energy demand rising fast with population + industrialisation. India imports ~85% of its crude oil.
- Renewables are **decentralised** – good for remote villages where the grid cannot reach.
- Energy security, rural employment, low running cost.

❖Potential in India

- Solar: ~5,000 trillion kWh/year incident on land; ~**300 clear sunny days** in most parts; 4–7 kWh/m^2/day.
- Wind: ~ 300 GW potential at 100 m hub height (Tamil Nadu, Gujarat, Karnataka, A.P., Maharashtra, Rajasthan).
- Biomass: ~ 25 GW (agri-residue, bagasse cogeneration), plus biogas from cattle dung.
- Small hydro: ~ 20 GW (< 25 MW stations).
- Geothermal: ~10,000 MW (Puga Valley – Ladakh, Manikaran – H.P., Tattapani – Chhattisgarh).
- National target: **500 GW non-fossil capacity by 2030**; Ministry: **MNRE**.

Renewable = inexhaustible + clean + local | Limitation = low energy density, intermittent, high initial cost

❖Comparison table

Feature	Conventional	Non-conventional
Availability	Limited	Unlimited
Pollution	High	Very low / nil
Initial cost	Moderate	High
Running cost	High (fuel)	Very low
Nature	Centralised	Mostly decentralised

2. The Solar Energy Option

- Sun = a giant fusion reactor, delivering ~ 1.8×10^{14} kW to earth; this is ~10,000 times mankind's present need.
- Free, widely distributed, clean, no moving fuel, long life of devices.
- **Forms of use:** (i) Thermal – heating, drying, cooking, power (ii) Photovoltaic – direct electricity (iii) Chemical – photosynthesis, solar hydrogen.
- **Disadvantages:** dilute (~1 kW/m^2 max), intermittent (night, clouds, season), needs storage/back-up, large land area, high capital cost.
- Solar radiation arrives as **short-wave** (0.3–3 µm) – ~7% UV, 47% visible, 46% IR.

3. Environmental Impact of Solar Power

❖Positive

- Zero emission during operation – no CO_2, SO_2, NO_x, no noise, no fuel transport.
- 1 kW rooftop PV saves ~ 1.2–1.5 tonnes CO_2 per year.
- Reduces dependence on fossil fuel; supports rural electrification.

❖Negative

- **Land use:** utility plants need ~ 2–3 hectares per MW; may displace habitat/agriculture.
- **Manufacturing:** PV cells use toxic chemicals (silicon tetrachloride, cadmium, arsenic, lead in solder); energy-intensive silicon purification.
- **Water use:** concentrating thermal plants need cooling water in arid zones.
- **Disposal:** end-of-life panels (25–30 yrs) → e-waste, recycling needed.
- **Heat island/ glare:** reflectors can harm birds (burn "streamers") and cause glare.

- Working fluids/ salts in solar ponds and storage may leak.

> Net environmental impact of solar is far lower than coal; energy pay-back time of a PV panel = 1–3 years

4. Physics of the Sun

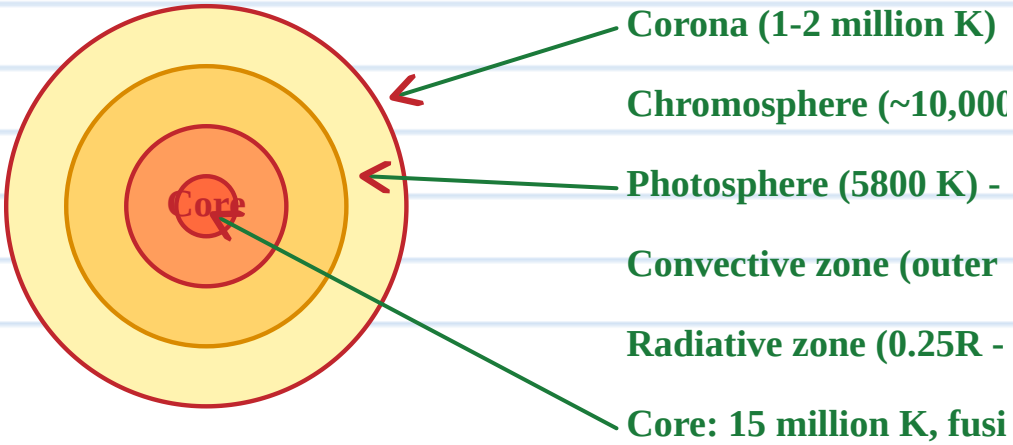


Fig 1.1 Structure of the Sun (not to scale)

- Diameter of sun = 1.39×10^9 m (about 109 × earth). Mass = 1.99×10^{30} kg. Mean distance from earth = 1.496×10^{11} m (1 AU).
- Effective black-body temperature = 5777 K (≈5800 K); core ≈ 15×10^6 K, density ≈ 150 g/cc.
- Made of ~ 73% H_2, 25% He, 2% heavier elements.

❖Energy source – nuclear fusion (proton-proton chain)

$$4\,{}^1H \rightarrow {}^4He + 2\,e^+ + 2\,\nu + 26.7\ MeV$$

- Mass lost ≈ 4.3×10^9 kg every second ($\Delta m \rightarrow E = mc^2$); total power output ≈ 3.8×10^{26} W.
- The energy diffuses outwards through the radiative and convective zones (takes ~ 10^5 yrs) and is emitted from the photosphere.

❖Spectral distribution

- Sun behaves like a black body at 5777 K – Wien's law: $\lambda_{max}\, T = 2898\ \mu m \cdot K$ → $\lambda_{max} \approx 0.50\ \mu m$ (green-yellow).
- Stefan–Boltzmann: $E = \sigma T^4$, $\sigma = 5.67 \times 10^{-8}\ W/m^2K^4$.

5. The Solar Constant (I_{sc})

> **Solar constant** = the energy from the sun, per unit time, received on a unit area of surface **perpendicular** to the rays, at the **mean earth-sun distance**, **outside the atmosphere.**

$$I_{sc} = 1367\ W/m^2\ (WMO\ value) = 1.94\ cal/cm^2/min = 4.92\ MJ/m^2h = 1.353\ kW/m^2\ (NASA)$$

❖Derivation (using Stefan-Boltzmann)

- Sun's power output $Q = 4\pi R_s^2 \sigma T_s^4$
- This is spread over a sphere of radius r (earth-sun distance), so
- $I_{sc} = (R_s / r)^2 \sigma T_s^4$
- With $R_s = 6.96\times10^8$ m, $r = 1.496\times10^{11}$ m, $T = 5777$ K → $I_{sc} \approx 1367\ W/m^2$.

❖Variation through the year

- Earth's orbit is elliptical → distance varies ±1.7%.
- Extra-terrestrial normal radiation on day n:

$$I_{on} = I_{sc}\,[\,1 + 0.033\cos(\,360\,n / 365\,)\,]$$

- Max ≈ 1412 W/m^2 (3 Jan, perihelion); Min ≈ 1322 W/m^2 (4 July, aphelion).

6. Extra-Terrestrial & Terrestrial Radiation

❖Extra-terrestrial radiation (ETR)

- Radiation falling on top of the atmosphere – undepleted; depends only on **latitude, day of the year, time** (since no atmosphere).
- Spectrum ≈ black-body 5777 K : UV 7%, visible 47%, IR 46%.
- On a **horizontal** surface at the top of atmosphere:

$$I_o = I_{on}\cos\theta_z = I_{on}\,(\sin\varphi\sin\delta + \cos\varphi\cos\delta\cos\omega)$$

- Daily total $H_o = (24/\pi)\,I_{sc}\,[1+0.033\cos(360n/365)]\,(\cos\varphi\cos\delta\sin\omega_s + (\pi\,\omega_s/180)\sin\varphi\sin\delta) \rightarrow J/m^2day$

❖Terrestrial radiation

- Radiation reaching earth's surface after passing the atmosphere; it is reduced and altered:
- Absorption by O_3 (UV), H_2O vapour & CO_2 (IR), dust.
- Scattering – by air molecules (Rayleigh, $\propto 1/\lambda^4$ → blue sky) and dust (Mie scattering).
- Reflection by clouds (albedo ≈ 30% on average for earth).
- Result: only ~ 70–75% reaches ground; peak ≈ 1000 W/m^2 at sea level on a clear day.

❖Components

- Beam (direct) radiation, I_b: received directly without scattering; casts shadow.
- Diffuse (sky) radiation, I_d: received after scattering by atmosphere; from all directions, no shadow.
- Global (total) radiation: $I = I_b \cos\theta + I_d$ (on a horizontal surface, $I = I_b \cos\theta_z + I_d$).
- Albedo/ reflected radiation from ground adds on tilted surfaces.

7. Sun–Earth Geometry & Angles

> Declination (δ): angle between sun's rays and the equatorial plane. Varies ±23.45° through the year.

$$\delta = 23.45 \sin[360\,(284 + n)/365]$$ (n = day number, 1 Jan = 1)

- δ = +23.45° on 21 June (summer solstice); 0° on 21 Mar & 23 Sep (equinox); −23.45° on 22 Dec.
- Latitude (φ): angular position north/south of equator (+ve north).
- Hour angle (ω): angular displacement of sun east/west of local meridian; 15° per hour; ω = 0 at solar noon, +ve afternoon, −ve morning.

$$\omega = 15\,(\text{Solar time} - 12)\ \text{degrees}$$

- Solar time = standard time ± 4 ($L_{st} - L_{loc}$) + E, where E = equation of time (min). 4 min per degree of longitude. For India, L_{st} = 82.5° E.
- Zenith angle θ_z, Altitude $\alpha = 90 - \theta_z$, Azimuth γ_s.

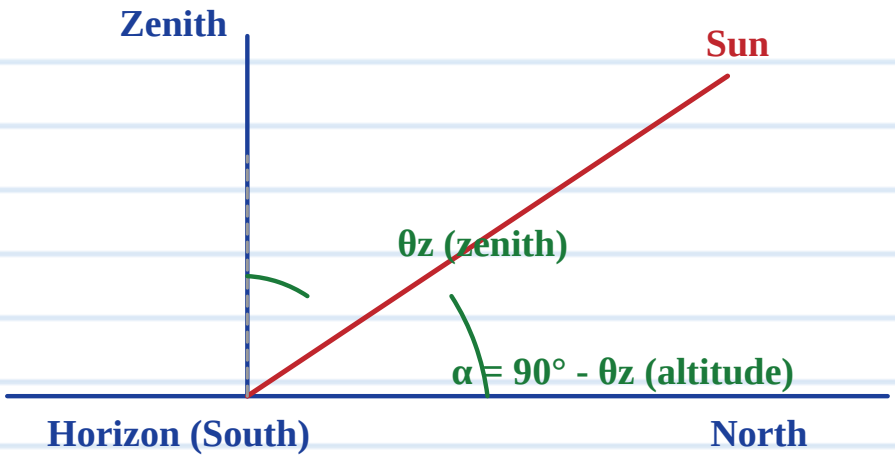


Fig 1.2 Zenith angle and solar altitude angle

$$\cos\theta_z = \sin\alpha = \sin\varphi \sin\delta + \cos\varphi \cos\delta \cos\omega$$

- Sunset hour angle: at sunset $\theta_z = 90° \rightarrow \cos\omega_s = -\tan\varphi \tan\delta$.
- Day length: $N = (2/15)\cos^{-1}(-\tan\varphi \tan\delta)$ hours.
- Surface azimuth (γ), slope (β), incidence angle (θ), and ground reflectance (ρ) are other terms.

❖Example

> Find the zenith angle at 10:00 solar time on 21 March at Hyderabad (φ = 17.4° N). $\delta = 0$, $\omega = 15(10-12) = -30°$. $\cos\theta_z = \cos17.4 \times \cos0 \times \cos(-30) = 0.954 \times 0.866 = 0.826 \rightarrow \theta_z = 34.3°$, altitude 55.7°.

8. Solar Radiation on a Tilted Surface

- Collectors are tilted (slope β) and face south (northern hemisphere), so we need radiation on the tilted plane from horizontal-plane data (I, I_d).

> Incidence angle θ = angle between the beam and the normal to the surface.

$$\cos\theta = \sin\delta \sin\varphi \cos\beta - \sin\delta \cos\varphi \sin\beta \cos\gamma + \cos\delta \cos\varphi \cos\beta \cos\omega + \cos\delta \sin\varphi \sin\beta \cos\gamma \cos\omega + \cos\delta \sin\beta \sin\gamma \sin\omega$$

- For a south-facing surface ($\gamma = 0$) this reduces to:

$$\cos\theta = \sin\delta \sin(\varphi - \beta) + \cos\delta \cos\omega \cos(\varphi - \beta)$$

- For a horizontal surface $\beta = 0 \rightarrow \cos\theta = \cos\theta_z$.
- For a surface that always faces the sun (tracking) $\rightarrow \theta = 0$.

❖Tilt factor (R_b)

$$R_b = I_{bt} / I_{bh} = \cos\theta / \cos\theta_z$$

❖Liu & Jordan isotropic-sky model

$$I_T = I_b R_b + I_d (1 + \cos\beta)/2 + \rho (I_b + I_d)(1 - \cos\beta)/2$$

- Term 1 = beam on tilted plane; Term 2 = diffuse from sky (sky view factor $(1+\cos\beta)/2$); Term 3 = ground-reflected ($\rho \approx 0.2$ for ground, 0.7 for snow).
- Best fixed tilt: $\beta \approx$ latitude (φ) for annual use; $\varphi + 15°$ for winter; $\varphi - 15°$ for summer.
- Monthly average: $\bar{R} = (1 - \bar{H_d}/\bar{H})\,\bar{R_b} + (\bar{H_d}/\bar{H})(1+\cos\beta)/2 + \rho(1-\cos\beta)/2$.

❖Example

Hyderabad ($\varphi = 17.4°$), noon ($\omega = 0$) on 21 June ($\delta = 23.45°$), south-facing surface tilted $\beta = 20°$. $\cos\theta = \sin 23.45 \sin(17.4-20) + \cos 23.45 \cos(-2.6) = 0.398\times(-0.0454) + 0.917\times0.999 = 0.898 \rightarrow \theta \approx 26.1°$.

9. Instruments for Measuring Solar Radiation

Radiometer → measures radiation (W/m^2). **Pyranometer** → global (beam+diffuse) on a plane. **Pyrheliometer** → beam (direct normal) only.

❖A. Pyranometer

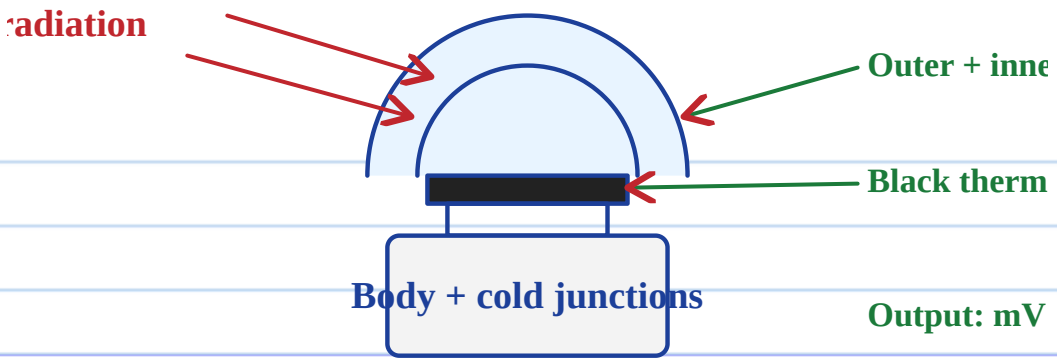


Fig 1.3 Pyranometer (thermopile type) – measures global (beam + diffuse) radiation

- **Thermopile type (Eppley / Kipp & Zonen):** a black, blackened receiver heated by radiation; temperature difference between hot (black) and cold (white) junctions → emf (mV) ∝ radiation. Covered by two hemispherical glass domes (0.3–3 μm transmission) to cut convective loss. Silica gel dryer inside.

- **Photovoltaic type:** silicon cell short-circuit current ∝ radiation – cheap but not spectrally flat.
- **Shadow-ring (shading band) method:** a band shades the pyranometer from direct sun → measures diffuse only. Correction factor applied.
- **Accuracy:** ± 1–3%; calibrated against a reference pyrheliometer.

❖B. Pyrheliometer (beam radiation)

- A long collimating tube with small field of view (~5.7°) → only direct sun disc + circumsolar region enters.
- Must be kept **pointed at the sun** with a **tracker** (equatorial mount).
- Types: **Angstrom compensation pyrheliometer** (two strips; one shaded & electrically heated until temperatures equal – electrical power = radiation), **Abbot silver-disc, Eppley normal incidence pyrheliometer (NIP).**
- Thermopile output → recorder.

❖C. Pyrgeometer / Net pyrradiometer

- Pyrgeometer measures long-wave (infrared) sky/earth radiation (4–100 μm).
- **Net pyrradiometer** – two blackened receivers facing up and down → net all-wave radiation.

10. Instruments for Sunshine Duration

> **Sunshine hours** = number of hours during which direct sunlight is strong enough (> 120 W/m^2 – WMO) to burn the record. **Bright sunshine** ≈ 0.2 kW/m^2 threshold.

❖Campbell–Stokes sunshine recorder

- A solid **glass sphere (10 cm)** acts as a lens, focusing sun rays on a **card strip** held in a spherical bowl behind it.
- The image burns a trace on the card as sun moves; **length of burn = sunshine duration.**
- Three different cards used: long curved (summer), short curved (equinox), straight (winter).
- Simple, cheap, no power. Limitation: thin cloud/ low sun is hard to record; observer needs to read the card daily.

❖ **Jordan sunshine recorder**

- Uses a semi-cylindrical box with two slits; light falls on **photosensitive paper** (blue-print) inside – the exposed line shows duration.

❖ **Modern: Pyrheliometric / photoelectric**

- Compares the beam signal with 120 W/m^2 threshold electronically → logs sunshine hours.

❖ **Uses of sunshine data**

- Gives **ratio n/N** (n = actual bright sun hours, N = max possible day length).
- Used in **Angstrom-Prescott equation** to estimate global radiation where no pyranometer is available:

$$\bar{H} / \bar{H_o} = a + b\,(\bar{n} / \bar{N})$$

- a, b = regression constants (typical $a \approx 0.25$, $b \approx 0.50$; for India often $a=0.27$, $b=0.47$).
- $\bar{H}$ = monthly mean daily global radiation on horizontal surface; $\bar{H_o}$ = extra-terrestrial.
- **Clearness index** $\bar{K_T} = \bar{H} / \bar{H_o}$.

II. Solar Radiation Data

- Needed for **designing and sizing** solar systems (collector area, PV array size, storage).
- **Sources:** (i) India Meteorological Department (IMD) network – ~45 stations measuring global + diffuse; (ii) **NREL / MNRE – Solar Radiation Handbook & Solar Resource Assessment (SRRA) stations;** (iii) NASA, PVGIS, Meteonorm (satellite derived).
- **Typical data given:** monthly average daily global radiation $\bar{H}$ (MJ)/m^2 day), diffuse $\bar{H_d}$, sunshine hours, ambient temperature.
- **Data types:** hourly, daily, monthly mean, **Typical Meteorological Year (TMY)** files.
- **Satellite methods** (INSAT, Kalpana) estimate radiation from cloud cover.

❖Average daily global radiation in India

Region	kWh/m²/day
Rajasthan, Gujarat (W. India)	5.5 – 6.5
Deccan plateau (A.P., Karnataka)	5.0 – 5.8
Eastern coast	4.5 – 5.5
North-East, Kerala (monsoon)	4.0 – 4.7

- India receives ~ 4–7 kWh/m²/day and 2300–3200 sunshine hours/yr.
- Units: 1 kWh/m² = 3.6 MJ/m²; 1 Langley = 1 cal/cm² = 41.87 kJ/m²; 1 W/m² = 0.86 kcal/m²h.

12. Solved Problems

❖Problem 1 – Day length

> Find the day length at $\varphi = 28^\circ$ N on 21 June. ($\delta = 23.45^\circ$)

$\cos \omega_s = -\tan 28^\circ \times \tan 23.45^\circ = -0.5317 \times 0.4336 = -0.2305 \rightarrow \omega_s = 103.3^\circ \rightarrow N = 2\times103.3/15 = 13.77$ h.

❖Problem 2 – Declination

> Declination on 15 April (n = 105): $\delta = 23.45 \sin[360(284+105)/365] = 23.45 \sin(383.7^\circ) = 23.45 \sin(23.7^\circ) = 9.4^\circ$.

❖Problem 3 – Extra-terrestrial normal radiation on 1 Feb (n = 32)

> $I_{on} = 1367[1 + 0.033 \cos(360\times32/365)] = 1367[1 + 0.033\times\cos 31.56^\circ] = 1367\times1.0281 = 1405$ W/m².

❖Problem 4 – Solar time

> Standard time 10:30 IST at Hyderabad (longitude 78.5° E) on a day when E = −10 min. Solar time = 10:30 + 4(78.5 − 82.5) min + E = 10:30 − 16 min − 10 min = 10:04.

❖Problem 5 – Angstrom

> $\bar{n}/N = 0.7$, $H_o = 36$ MJ/m²d, a = 0.25, b = 0.50 → $H = 36\,(0.25 + 0.5\times0.7) = 36 \times 0.60 = 21.6$ MJ/m²d.

13. Important Questions (Exam Oriented)

- Explain the role and potential of new & renewable energy sources.

- What are the environmental impacts of solar power?
- Define the solar constant; derive it, and explain the variation of ETR through the year.
- Define declination, hour angle, zenith angle, incidence angle. Derive $\cos\theta_z$.
- Explain terrestrial solar radiation; beam, diffuse and global components.
- Derive the expression for radiation on a tilted surface (Liu & Jordan).
- With a neat sketch, explain the pyranometer and pyrheliometer.
- Explain the Campbell–Stokes sunshine recorder.
- Write notes on solar radiation data & Angstrom correlation.

Tip: draw neat labelled diagrams, write equations clearly with units – full marks!

14. Two More Problems

❖Problem 6 – Sunset hour angle & sunrise time

At $\varphi = 17.4°$ N on 22 Dec ($\delta = -23.45°$): $\cos\omega_s = -\tan 17.4 \times \tan(-23.45) = +0.3134 \times 0.4336 = 0.1359 \rightarrow \omega_s = 82.2° \rightarrow$ day length $= 2 \times 82.2/15 = 10.96$ h; sunrise ≈ 06:31 solar time, sunset ≈ 17:29 solar time.

❖Problem 7 – Radiation on tilted plane

$I_b = 600$, $I_d = 120$ W/m² (horizontal basis, $I_{bh} = 600$), $R_b = 1.2$, $\beta = 30°$, $\rho = 0.2$.

$I_T = 600(1.2) + 120(1+0.866)/2 + 0.2(720)(1-0.866)/2 = 720 + 112 + 9.6 = 841.6$ W/m².

15. Quick Revision – Formula Sheet

Quantity	Formula
Solar constant	$I_{sc} = 1367\ W/m^2$
ETR on day n	$I_{on} = I_{sc}[1+0.033\cos(360n/365)]$
Declination	$\delta = 23.45 \sin[360(284+n)/365]$
Hour angle	$\omega = 15(t_{solar} - 12)$
Zenith	$\cos\theta_z = \sin\varphi \sin\delta + \cos\varphi \cos\delta \cos\omega$
Sunset angle	$\cos\omega_s = -\tan\varphi \tan\delta$
Day length	$N = (2/15)\,\omega_s$ hours
Tilted incidence ($\gamma=0$)	$\cos\theta = \sin\delta \sin(\varphi-\beta) + \cos\delta \cos\omega \cos(\varphi-\beta)$
Tilt factor	$R_b = \cos\theta / \cos\theta_z$
Angstrom	$H/H_o = a + b(n/N)$

❖Key terms in one line each

- **Air mass (m):** ratio of path length of beam through atmosphere to the path when sun is at zenith; $m = 1/\cos\theta_z = 1$ at zenith, AM 1.5 standard for PV testing.
- **Sky clearness / Turbidity:** measure of dust and water vapour in atmosphere.
- **Insolation:** incident solar radiation energy per unit area (Wh/m^2).
  Irradiance: power per unit area (W/m^2).